Information Science Lab

EWSHM 2026
12th European Workshop on Structural Health Monitoring

Hybrid Event-Driven Structural Health Monitoring (EDSHM) Node with Ultra-Low Power Adaptive Triggering

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Students & Young Professionals Challenge

Motivation: Why Current Wireless SHM is Not Sustainable

Current Wireless SHM

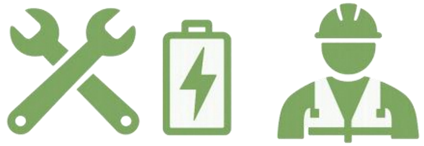
- Continuous Sensing
- Idle Listening
- Periodic Transmission

Consequence

High Energy Consumption

The battery drains rapidly due to unnecessary operation

Impact



Frequent Maintenance

Battery replacement and increased operational cost

Conventional wireless SHM continuously consumes energy even when no meaningful structural events occur

Innovations and System-Level Benefits

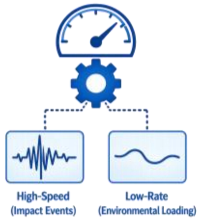
1 Zero-Power Vibrating Triggering

- A piezoelectric sensor used as a passive event detector
- Hardware wake-up through energy threshold detection
- Ultra-low standby power

2 Gateway-Assisted Predictive Monitoring

- Weather forecasts analyzed at the gateway
- Wake-up radio activation during high-risk conditions
- No idle listening or periodic weather checks

3 Context-Aware Adaptive Sensing



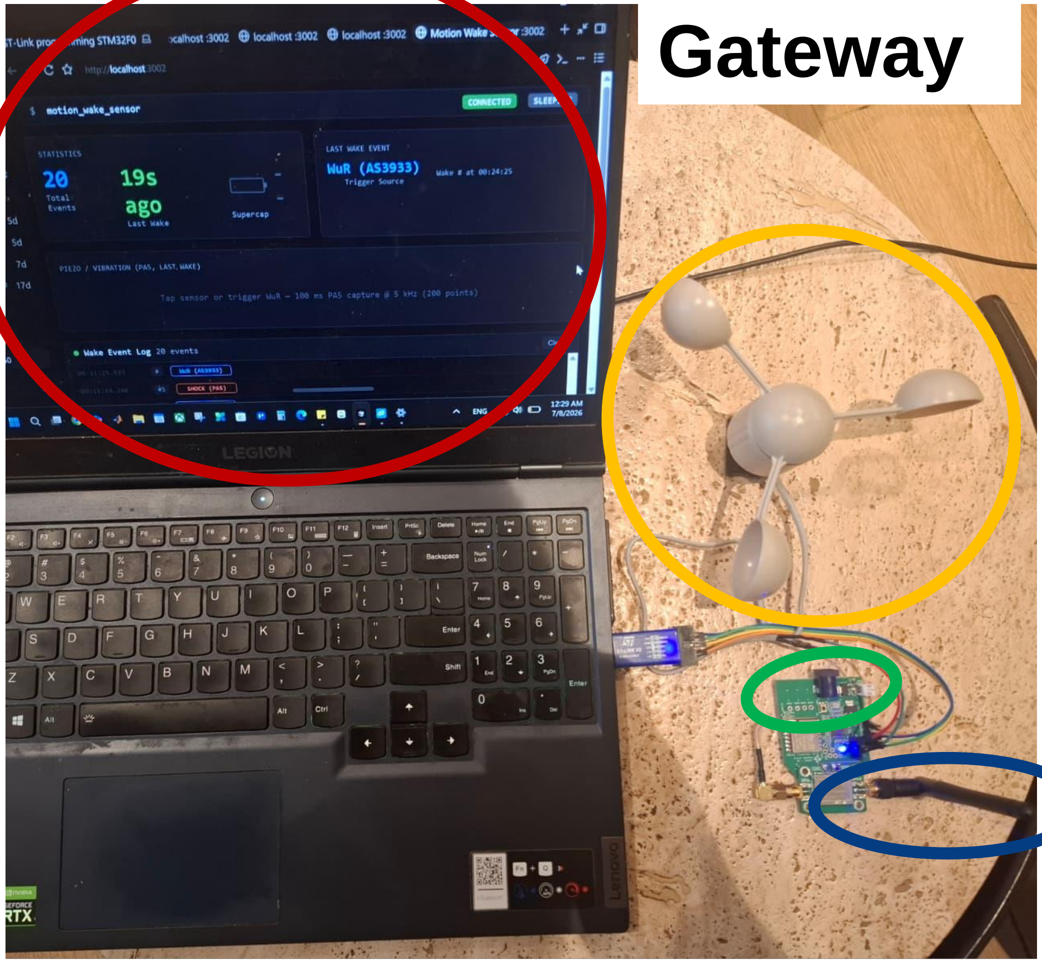
- High-speed acquisition for impact events
- Low-rate monitoring for environmental loading
- Intelligent selection of sensing mode

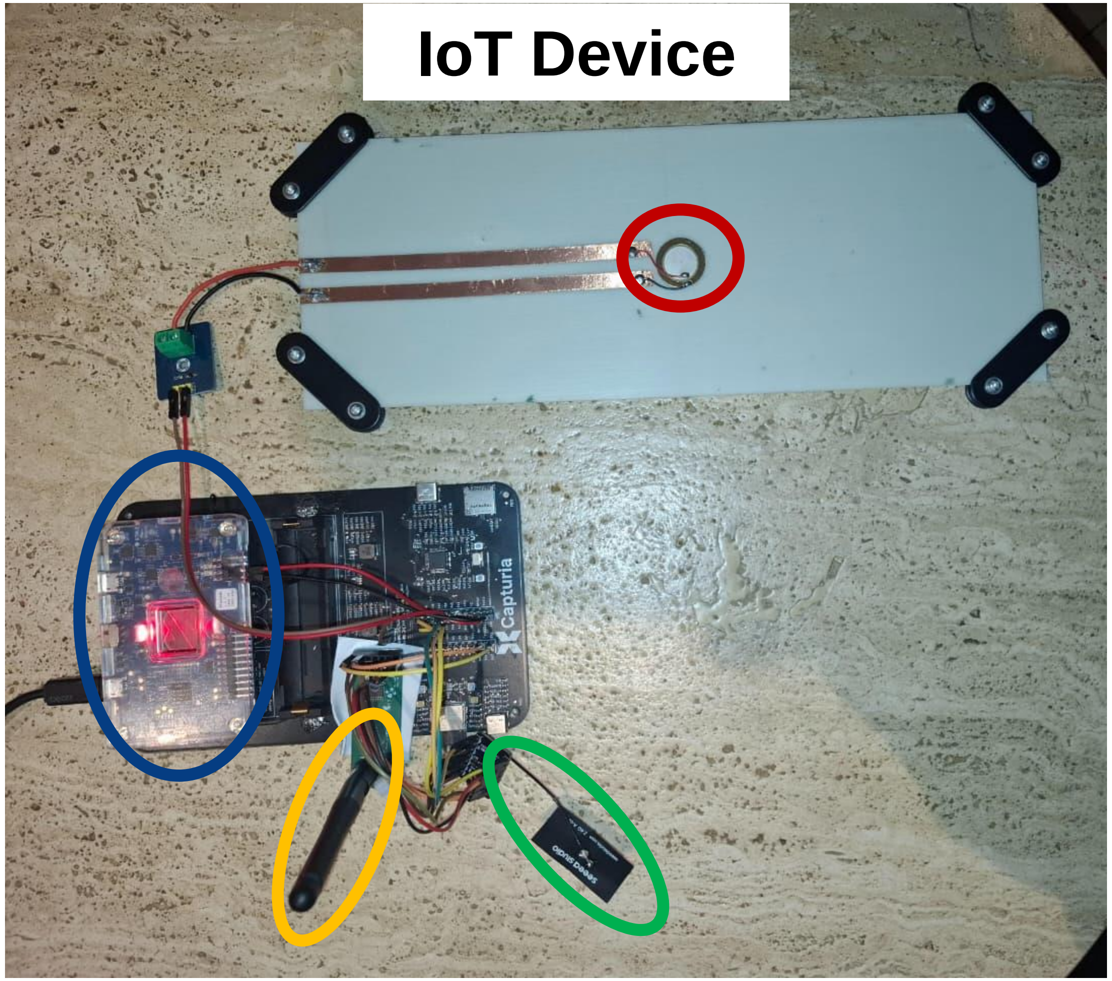
Event-Driven SHM

- Reduced Power Consumption
- Reduced Data Transmission
- Extended Operational Lifetime
- Scalable Wireless SHM Deployment

"We shift wireless SHM from periodic monitoring to intelligent event-driven monitoring, enabling long-term autonomous sensing with minimal energy consumption."

Proposed Solution: Hybrid Event-Driven SHM Architecture

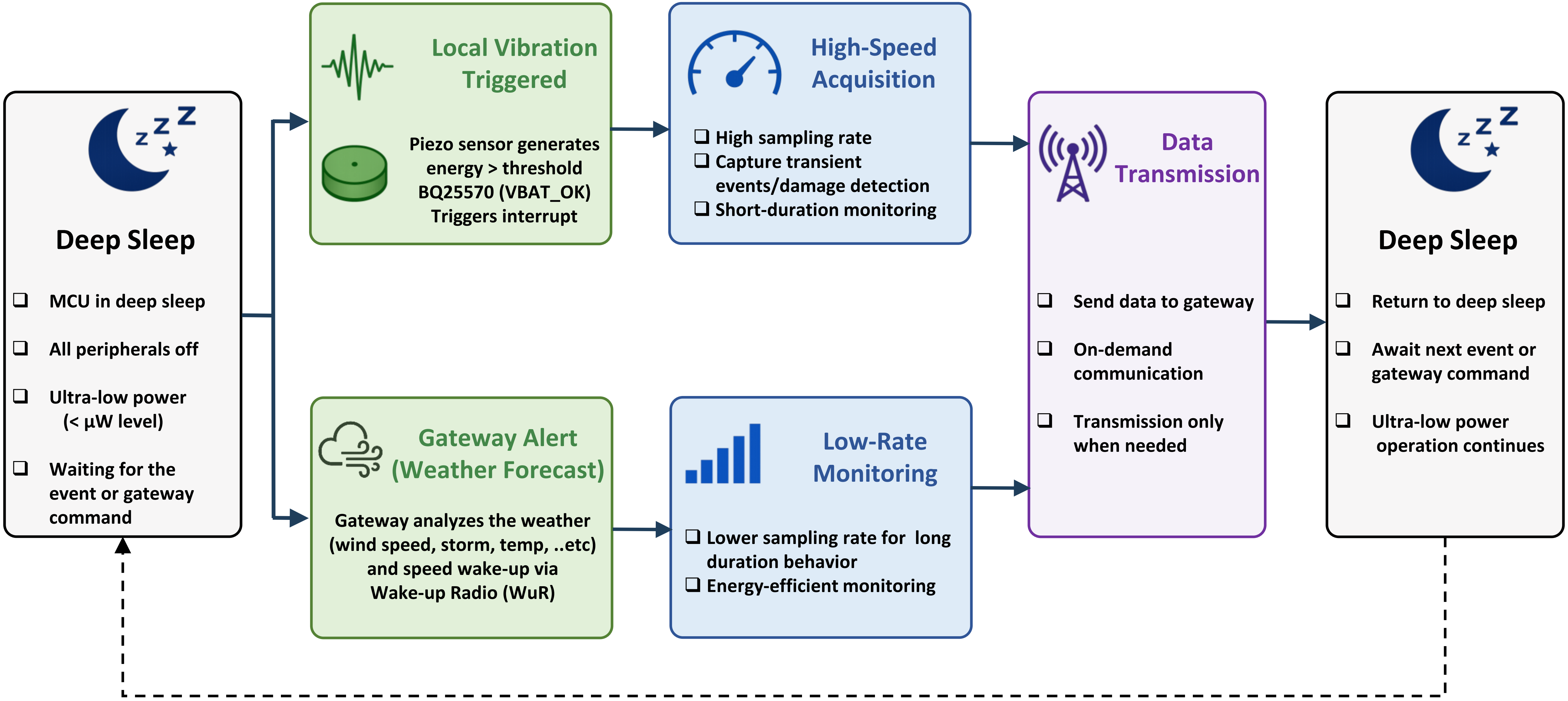




Central Dashboard (Data and Events Logging)
Gateway Microcontroller
Wind Sensor (Mock-up for weather station)
Wake-up transmitter

Sensor
ESP32 : Wireless Data Transmission
Wake-up Radio : Wireless Activation
Power Profiler Kit

How It Works: Operational Workflow of the Proposed EDSHM Node

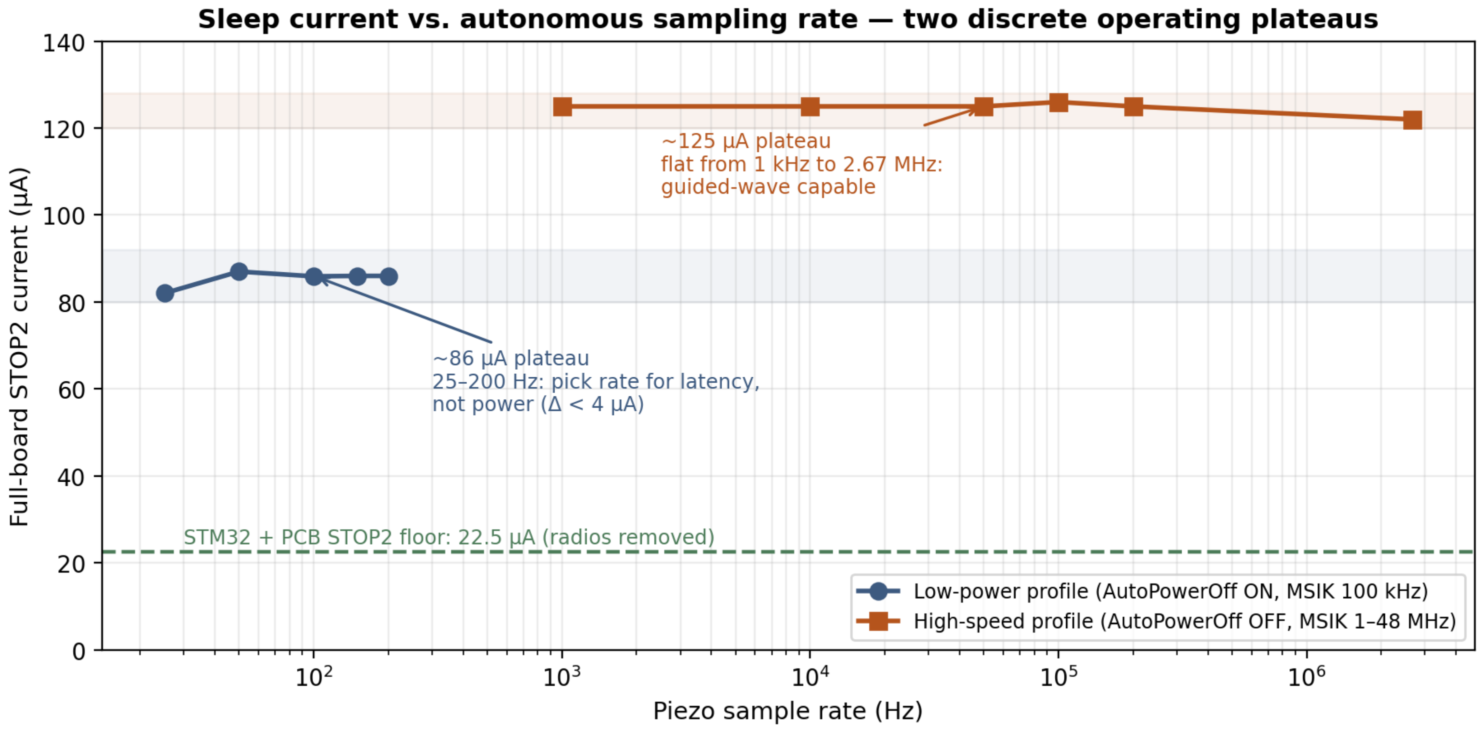


Cycle repeats for continuous, ultra-low-power monitoring

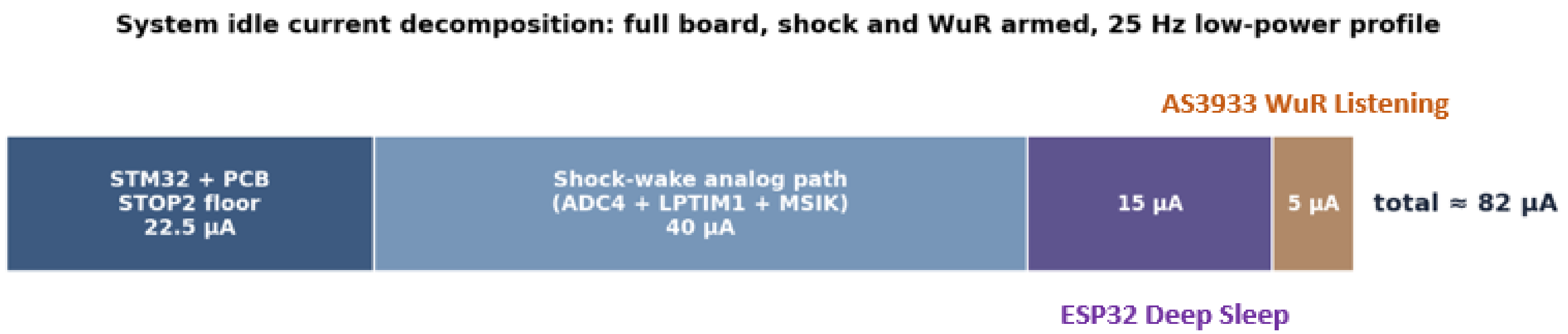
Why It Is Better: Power Consumption

Autonomous Sensing: Profiling Rate versus Sleep Current

Operating point	Configuration	Sleep current	Usable sample rates
Low-power monitoring	MSIK 100 kHz, LPTIM ÷64, ADC ÷32, AutoPowerOff ON	82 to 90 µA	25 to 200 Hz (Δ < 4 µA across the range)
High-speed capture	MSIK 1 to 48 MHz, LPTIM ÷1, ADC ÷1, AutoPowerOff OFF	122 to 126 µA	1 kHz to 2.67 MHz (flat across the range)



Why It Is Better: Power Consumption



Intervention	Idle current after	Class
Baseline, as received	about 3.4 mA	
Mask ADC EOC interrupt; AWD-only wake	about 2.55 mA	firmware (−850 µA)
Remove I2C monitors (MAX17048, INA226): no full-disable provision, residual standby draw	about 2.37 mA	hardware (−180 µA)
Remove Y1 programmable oscillator: E/D pin floats high by internal pull-up; no shunt, no GPIO control exists	about 25 µA	hardware (−2.35 mA, root cause)
Full node: shock path armed at 25 Hz, WuR armed, ESP32 in deep sleep	about 82 µA	operational point
STM32 and PCB floor (radios removed, all wake peripherals off)	22.5 µA	benchmark

Why It Is Better: Energy Consumption and Battery Lifetime Comparison

Configuration	Listen mode	Cycle period (s)	Avg current (µA)	Battery lifetime (yr)	Responsiveness — worst-case latency	P(miss event)	Miss type / note
DCY — 1 s cycle	wake every 1 s	1	126.71	0.38	<= 1 s	0.0%	*Timing miss (event lost)*
DCY — 12 s cycle	wake every 12 s	12	11.34	4.23	<= 12 s	3.3%	*Timing miss (event lost)*
DCY — 44 s cycle	wake every 44 s	44	3.66	13.10	<= 44 s	73.6%	*Timing miss (event lost)*
DCY — 60 s cycle (1 min)	wake every 60 s	60	2.89	16.58	<= 60 s (avg 30 s)	80.7%	*Timing miss (event lost)*
WuRIoT (ours)	Wu-Rx always on	—	**3.68**	**13.03**	**~ ms (near-instant)**	**3.2%**	*Channel miss (recoverable next pass)*

WuRIoT breaks the fundamental trade-off that duty-cycling can't escape — delivering a 13-year battery life and near-instant, ~97% reliable event capture, where a duty-cycled device must sacrifice one to gain the other.

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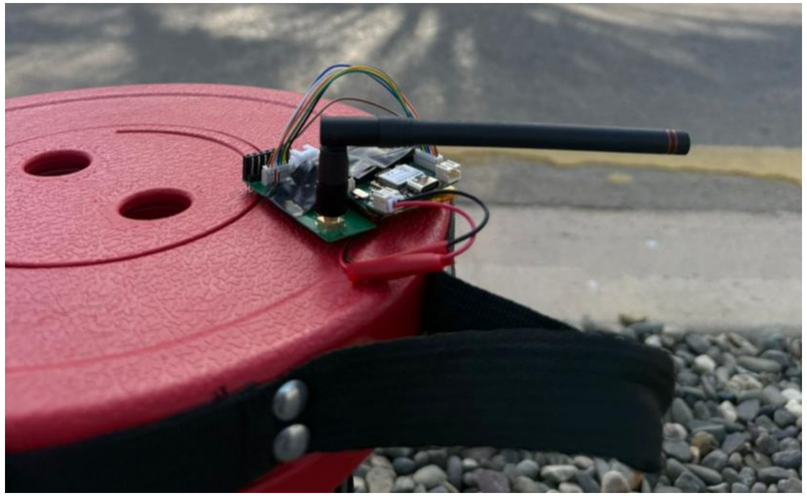
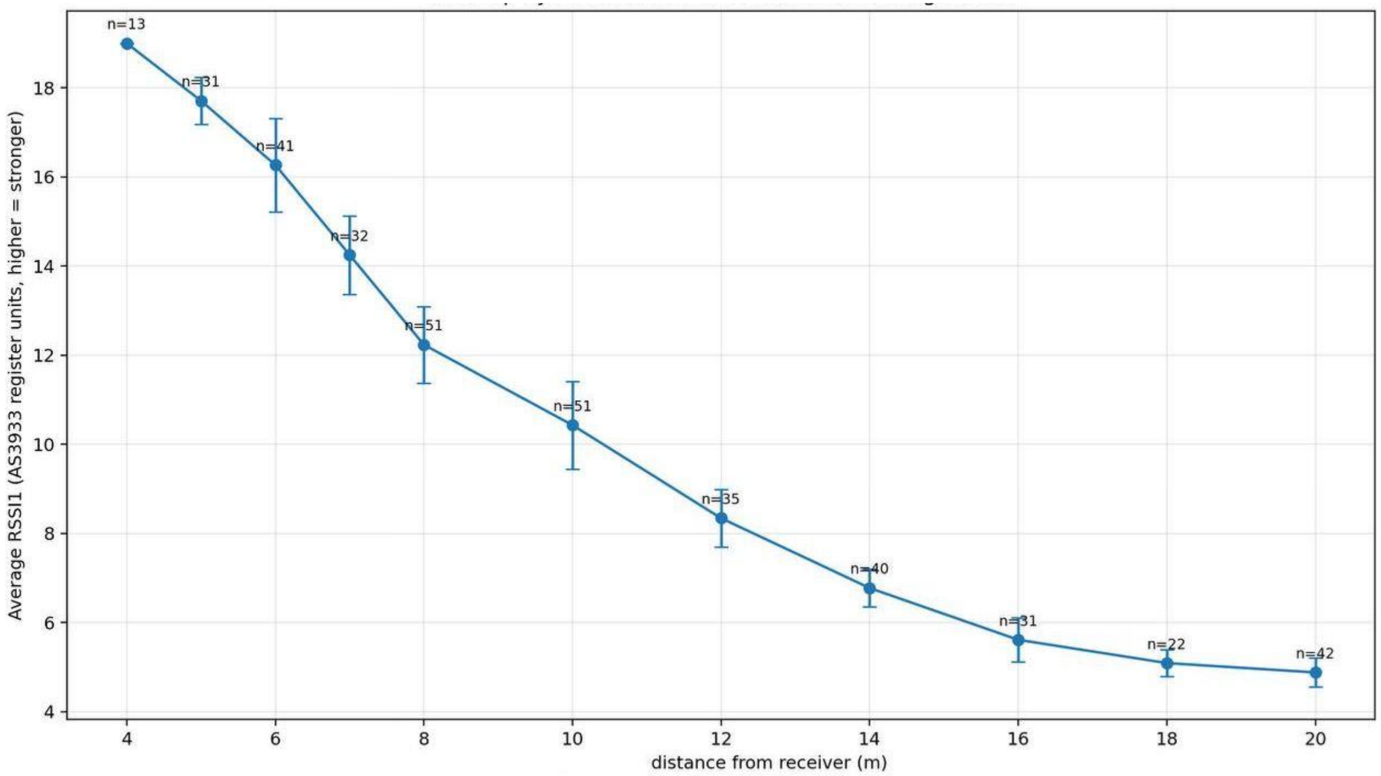
Use Case Example





Thank you!